Strong disorder and critical-point shift for a Gaussian pinning model

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Abstract

We study renewal pinning with independent standard Gaussian disorder and gap law $K(n) = n^{-5/4}/\zeta(5/4)$. For every inverse temperature $\beta > 0$, we prove that almost-sure extinction of the normalized free partition function at the annealed field $-\beta^2/2$ implies a strictly larger quenched critical field. The decreasing renewal hazard allows us to control future intersections through the polymer's age distribution. Under strong disorder, this gives a $1/t$ lower tail for the maximum constrained partition function. Testing favorable stretches along a planted renewal, while retaining the final renewal endpoint, then gives summable fractional moments and a positive change of field with zero free energy. A quantitative estimate of Alexander and Berger excludes extinction at and below the second-moment boundary.

1 Introduction

In a renewal pinning model, the contact times of a polymer with a defect line form a renewal process. Random rewards at the contacts compete with the entropy of long excursions. The critical field marks the onset of positive free energy, while the normalized partition function at the annealed field provides a martingale test for disorder. These two descriptions of the transition need not coincide for a general disordered system.

We consider the following fixed renewal law. Let $\tau = \{0 = \tau_0 < \tau_1 < \cdots\}$ have iid gaps with

$$K(n) := P_\tau(\tau_1 = n) = \frac{n^{-5/4}}{\zeta(5/4)}, \qquad n \geq 1, \quad \zeta(5/4) = \sum_{m \geq 1} m^{-5/4}. \tag{1.1}$$

Let $(\omega_n)_{n\geq 1}$ be iid standard real Gaussians, independent of τ, and write $\mathbb{P}, \mathbb{E}$ for their law and expectation. For $\beta > 0$ and $h \in \mathbb{R}$, define the free partition function

$$Z_{N,h}^{\beta,\omega} = E_\tau \exp\left(\sum_{n=1}^{N} (\beta\omega_n + h) 1_{\{n\in\tau\}}\right), \qquad F(\beta, h) = \lim_{N\to\infty} \frac{1}{N} \mathbb{E} \log Z_{N,h}^{\beta,\omega}, \tag{1.2}$$

and $h_c(\beta) = \inf\{h : F(\beta, h) > 0\}$. The endpoint at N is not constrained to be a contact. The annealed critical field is $h_a(\beta) = -\beta^2/2$. The process $W_N = Z_{N,h_a}^{\beta,\omega}$ is a nonnegative mean-one martingale, as follows by averaging the next Gaussian contact weight. Write W_∞ for its almost-sure limit. We call $\mathbb{P}(W_\infty = 0) = 1$ strong disorder.

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The martingale approach establishes disorder irrelevance in a small-disorder range for renewal exponents below 1/2; see Lacoin [4]. Fractional-moment estimates give a strict critical-point shift for sufficiently large Gaussian disorder [5]. Neither statement alone identifies what happens at every inverse temperature for which the martingale becomes extinct. Berger [2, Conjecture 4.9] asks whether strong disorder forces a strict critical-point shift for pinning laws with nonzero renewal exponent. Our result treats the exact law (1.1).

Theorem 1.1. *For the renewal law (1.1) and independent standard Gaussian disorder, for every $\beta > 0$,*

$$\mathbb{P}(W_\infty = 0) = 1 \quad \Longrightarrow \quad h_c(\beta) > -\beta^2/2.$$

More precisely, under this premise there is $\Delta_\beta > 0$ such that $F(\beta, -\beta^2/2 + \Delta_\beta) = 0$.

Thus extinction at the annealed field is accompanied by a genuine interval of zero free energy above that field. The positive interval may depend on β. The theorem uses the exact normalized power law; it makes no claim for arbitrary slowly varying modifications of its tail.

The proof follows the strategy of finding large partition weights and detecting them under size bias, as used for bulk directed polymers by Junk and Lacoin [3]. In renewal pinning, however, the disorder is collected only at contacts, and long excursions have polynomial probabilities. To control where the polymer can collect disorder, we use its age since the last contact. The decreasing hazard in (1.1) makes the squared future contact probabilities uniformly small when the next contact probability is small. Above the second-moment boundary, their conditional drift forces a large constrained weight before a high martingale level is reached.

Such weights can be found in logarithmic windows. Under a Gaussian size bias, the environment contains a planted renewal. Tests begun at its renewal times have iid failure-and-length pairs. Keeping the final planted endpoint in this calculation preserves the power $K(N)$; this is what makes the resulting fractional moments summable and permits a strictly positive change of field. The second-moment boundary itself is handled separately, using the quantitative coarse-graining estimate for one renewal pinned on another in Alexander and Berger [1].

Section 2 develops the age estimate, and Section 3 proves the large-weight bound. Section 4 obtains the positive field change. Section 5 treats the remaining boundary and completes the proof. Constants may depend on the fixed β and on explicitly fixed auxiliary exponents. We suppress β, ω from partition functions when no ambiguity is possible.

2 The age process and future intersections

Throughout the proof, $s = 5/4$, $\alpha = s - 1 = 1/4$ and $c = \zeta(s)^{-1}$, so $K(n) = cn^{-s}$. Write

$$\eta_n = e^{\beta\omega_n - \beta^2/2}, \qquad \xi_n = \eta_n - 1, \qquad \chi = \mathbb{E}\xi_n^2 = e^{\beta^2} - 1, \qquad \mathcal{F}_n = \sigma(\omega_1, \dots, \omega_n).$$

All absolute moments of ξ_n are finite and $\xi_n > -1$. Let C_n be the partition function constrained to have a contact at n, at field $h_a = -\beta^2/2$. Then

$$C_0 = 1, \qquad C_n = \eta_n \sum_{j=0}^{n-1} K(n-j) C_j, \qquad W_N = \sum_{j=0}^{N} C_j \overline{K}(N-j), \quad \overline{K}(a) = \sum_{m>a} K(m). \tag{2.1}$$

In particular, $C_n \le W_n$, $\mathbb{E}C_n = u(n) := P_\tau(n \in \tau)$, and $u(0) = \overline{K}(0) = 1$. The renewal theorem [2, Theorem 3.16] gives $u(n) \asymp n^{-3/4}$. Hence

$$0 < R := \sum_{n \ge 1} u(n)^2 < \infty, \qquad \beta_2^2 := \log(1 + 1/R). \tag{2.2}$$

The strict inequality $\beta > \beta_2$ is equivalent to $\chi R > 1$. We first prove the critical-point shift in that range under strong disorder. Section 5 will exclude strong disorder at and below β_2.

The age of a renewal is the time elapsed since its last contact. Its transition matrix T on $\mathbb{N}_0$ resets age a to zero with probability

$$r(a) = \frac{K(a+1)}{\overline{K}(a)}$$

and otherwise moves to $a+1$. The following property of the exact law will control contacts far into the future.

Lemma 2.1. *The hazard $r(a)$ decreases to zero and satisfies $r(a) \le \alpha/(a+1)$. For $u_a(k) := T^k(a,0)$, $k \ge 1$, one has*

$$0 \le u_a(k) \le u(k), \qquad \sum_{k\ge 1} u_a(k)^2 \longrightarrow 0 \quad (a \to \infty).$$

Proof. With $m = a+1$,

$$r(a)^{-1} = \sum_{j\ge 0} \left(\frac{m}{m+j}\right)^s.$$

Each summand increases with m. Also $\sum_{n=m}^{\infty} n^{-s} \ge \int_m^{\infty} x^{-s}\,dx$, which gives the asserted upper bound on r. Couple chains begun at ages a and zero with the same uniform variables. The larger age remains larger: whenever it resets, the smaller age resets as well, and the chains then agree. Thus $u_a(k) \le u_0(k) = u(k)$. For fixed k, a reset before time k has probability at most $kr(a)$, so $u_a(k) \to 0$. Dominated summation using (2.2) proves the last claim. □

For a probability mass μ on ages, put

$$f_\mu(k) = (\mu T^k)(0), \quad k \ge 1, \qquad U(\mu) = \sum_{k\ge 1} f_\mu(k)^2. \tag{2.3}$$

This quantity measures the expected future overlap of two independent renewals begun with age law μ. We use the same linear definition of f_μ for differences of the probability masses appearing below. By lemma 2.1, $\|f_{\delta_a}\|_2 \le \sqrt{R}$, so these sums are well defined and $0 \le U(\mu) \le R$.

Lemma 2.2. *If $q = (\mu T)(0)$ tends to zero, then $U(\mu)$ and $U(\mu T)$ tend to zero uniformly over all probability masses μ. Moreover,*

$$U(\mu T) = U(\mu) - q^2. \tag{2.4}$$

Proof. For every integer $A \ge 0$, monotonicity of the hazard gives

$$\mu([0,A]) \le q/r(A), \qquad \|f_\mu\|_2 \le \sqrt{R}\, q/r(A) + \sup_{a>A} \|f_{\delta_a}\|_2.$$

Choose A large by lemma 2.1, and then choose q small. This proves uniform convergence. Finally $f_{\mu T}(k) = f_\mu(k+1)$, giving (2.4) and the assertion for μT. □

Under the polymer measure at time n, the age law is

$$\mu_n(a) = \frac{C_{n-a}\overline{K}(a)}{W_n}, \qquad 0 \le a \le n.$$

Given the past, let $v = \mu_{n-1}T$, $q_n = v(0)$ and $z = q_n\xi_n/(1+q_n\xi_n)$. Prediction by T, followed by reweighting the mass at zero by η_n, gives the exact identities

$$\frac{W_n}{W_{n-1}} = 1 + q_n\xi_n, \qquad \mu_n = v + z(\delta_0 - v). \tag{2.5}$$

Here $0 < q_n \le 1/4$ and q_n is $\mathcal{F}_{n-1}$-measurable. Let $\Delta U_n = U(\mu_n) - U(\mu_{n-1})$. Expanding the square in (2.3) gives

$$\Delta U_n = -q_n^2 + 2zH + z^2 J, \quad H = \langle f_v, f_{\delta_0 - v}\rangle, \quad J = \|f_{\delta_0 - v}\|_2^2. \tag{2.6}$$

Lemma 2.3. *Suppose $\beta > \beta_2$. Set $\gamma = (\chi R - 1)/2 > 0$. For every $\lambda > 0$ there is $\delta > 0$ such that, whenever $q_n \le \delta$,*

$$\mathbb{E}[e^{-\lambda \Delta U_n} \mid \mathcal{F}_{n-1}] \le e^{-aq_n^2}, \qquad a = \lambda\gamma/2. \tag{2.7}$$

Thus a can be made arbitrarily large, with a corresponding smaller δ.

Proof. For $q = q_n \le 1/4$, the denominator $1 + q\xi_n$ is at least $3/4$. Conditionally on the past,

$$|z| \le \tfrac{4}{3}q|\xi_n|, \quad |z| \le 1, \quad \mathbb{E}z = -q^2\mathbb{E}\frac{\xi_n^2}{1+q\xi_n} = O(q^2), \quad \mathbb{E}z^2 = \chi q^2 + O(q^3), \quad \mathbb{E}|z|^4 = O(q^4).$$

The error in the second moment follows by bounding $|(1+x)^{-2} - 1| \le C|x|$ for $x \ge -1/4$. By lemma 2.2, uniformly as $q \downarrow 0$,

$$|H| \le \sqrt{U(v)}(\sqrt{R} + \sqrt{U(v)}) \longrightarrow 0, \qquad J \longrightarrow R, \qquad 0 \le J \le 4R.$$

Equation (2.6) therefore implies

$$\frac{\mathbb{E}[\Delta U_n \mid \mathcal{F}_{n-1}]}{q^2} \longrightarrow \chi R - 1, \qquad \frac{\mathbb{E}[(\Delta U_n)^2 \mid \mathcal{F}_{n-1}]}{q^2} \longrightarrow 0. \tag{2.8}$$

For the latter assertion use $(\Delta U_n)^2 \le 3q^4 + 12H^2z^2 + 3J^2z^4$. Also $\Delta U_n \ge -q^2 - 2|H| \ge -b(\delta)$ for $q \le \delta$, where $b(\delta) \to 0$. For $x \ge -b$ the exponential remainder satisfies

$$e^{-\lambda x} \le 1 - \lambda x + \tfrac{1}{2}\lambda^2 e^{\lambda b}x^2.$$

Choose δ so the first conditional mean in (2.8) is at least γq^2 and the remainder contributes at most $\lambda\gamma q^2/2$. Then the conditional expectation is at most $1 - aq^2 \le e^{-aq^2}$. □

In particular, before a step with $q_n > \delta$,

$$\exp\left\{-\lambda U(\mu_n) + a\sum_{i\le n} q_i^2\right\} \tag{2.9}$$

is a nonnegative supermartingale. This stopping is legitimate before the exceptional step because q_n is known at time $n-1$.

3 Large constrained weights

In this section assume strong disorder, $\mathbb{P}(W_\infty = 0) = 1$, and $\beta > \beta_2$. The aim is a $1/t$ lower bound for the tail of $\sup_n C_n$. This will produce detectable favorable stretches whose length is only logarithmic in their weight.

Lemma 3.1. *Let $\tau_x = \inf\{n \geq 0 : W_n \geq x\}$, with $\inf \varnothing = \infty$. For every fixed $p \geq 1$ there is $M_p < \infty$ such that, for $x \geq 1$,*

$$\frac{1}{M_1 x} \leq \mathbb{P}(\tau_x < \infty) \leq \frac{1}{x}, \qquad \mathbb{E}[W_{\tau_x}^p; \tau_x < \infty] \leq M_p x^{p-1}. \tag{3.1}$$

Proof. First, lognormal relative overshoots have uniformly bounded moments:

$$\sup_{t \geq 1} \mathbb{E}[(\eta_1/t)^p \mid \eta_1 \geq t] < \infty. \tag{3.2}$$

Indeed, with $b = (\log t + \beta^2/2)/\beta \geq \beta/2$, the conditional expectation equals

$$\frac{\int_0^\infty e^{p\beta y - by - y^2/2}\, dy}{\int_0^\infty e^{-by-y^2/2}\, dy}.$$

For $b \geq \max(1, 2p\beta)$ the numerator is at most $(b - p\beta)^{-1}$ and the denominator is at least $e^{-3/2}/b$, by restricting to $[0, 1/b]$. On the remaining bounded interval, the numerator is uniformly finite and the denominator is bounded away from zero.

At a first crossing of $x > 1$, conditional on the past, the condition is $\eta_n \geq t = (x/W_{n-1} - 1 + q_n)/q_n > 1$. By (2.5),

$$\frac{W_n}{x} = \frac{1 - q_n + q_n\eta_n}{1 - q_n + q_n t} \leq \eta_n/t.$$

Apply (3.2) conditionally and sum over the disjoint first-crossing times. This gives

$$\mathbb{E}[W_{\tau_x}^p; \tau_x < \infty] \leq M_p x^p \mathbb{P}(\tau_x < \infty).$$

The nonnegative martingale maximal inequality gives the upper probability bound in (3.1). Bounded optional stopping yields

$$1 = \mathbb{E}[W_{\tau_x}; \tau_x \leq n] + \mathbb{E}[W_n; \tau_x > n].$$

The second term tends to zero by strong disorder and domination by x. Thus $\mathbb{E}[W_{\tau_x}; \tau_x < \infty] = 1$, proving the lower probability bound and then the moment bound. For $x = 1$, $\tau_1 = 0$; enlarge the constants to include this case. □

Proposition 3.2. *There is $c_\beta > 0$ such that*

$$\mathbb{P}\left(\sup_{n \geq 0} C_n \geq t\right) \geq c_\beta/t, \qquad t \geq 1. \tag{3.3}$$

Proof. Expanding the fourth power in (2.5), using $\mathbb{E}\xi_n = 0$, gives a constant $D > 0$ with

$$\mathbb{E}(1 + q\xi_n)^4 \leq e^{Dq^2} \quad (0 \leq q \leq 1/4).$$

Hence $W_n^4 \exp(-D\sum_{i\le n} q_i^2)$ is a nonnegative supermartingale. Choose $\varepsilon_0 > 0$ so $4 - D\varepsilon_0 \ge 3$. Choose λ, then δ, in lemma 2.3 so that $a\varepsilon_0 \ge 3$.

Let $B > 1$ be a level multiplier, to be fixed below. On the event that $\tau_{Bx} < \infty$, write $A = \sum_{n=\tau_x+1}^{\tau_{Bx}} q_n^2$. The sum is empty if a single overshoot crosses both levels. Stopping the fourth-moment supermartingale after τ_x and using lemma 3.1 gives

$$\mathbb{P}(\tau_{Bx} < \infty, A \le \varepsilon_0 \log B) \le M_4 x^{-1} B^{-4+D\varepsilon_0} \le M_4 x^{-1} B^{-3}. \tag{3.4}$$

At the terminal crossing, its fourth-power factor is at least $(Bx)^4$ and its compensator at least $B^{-D\varepsilon_0}$, whereas the initial expected fourth power is at most $M_4 x^3$.

Similarly (2.9) and $0 \le U \le R$ imply

$$\mathbb{P}\left(\tau_{Bx} < \infty, A \ge \varepsilon_0 \log B, q_n \le \delta \ \forall\, \tau_x < n \le \tau_{Bx}\right) \le x^{-1} e^{\lambda R} B^{-a\varepsilon_0} \le x^{-1} e^{\lambda R} B^{-3}. \tag{3.5}$$

To obtain this bound, condition at τ_x and stop before the first $q_n > \delta$; the terminal exponential on the indicated event is at least $e^{-\lambda R} B^{a\varepsilon_0}$ times its initial value. Multiply the conditional probability bound by $\mathbb{P}(\tau_x < \infty) \le 1/x$. Both stopping estimates follow first at bounded horizons and then by Fatou.

Let $\sigma = \inf\{n \ge \tau_x : W_n \le x/B\}$. Conditional on a dip before the upper hit, the probability of a later passage to Bx is at most $W_\sigma/(Bx) \le B^{-2}$. Therefore

$$\mathbb{P}(\sigma < \tau_{Bx} < \infty) \le x^{-1} B^{-2}. \tag{3.6}$$

By (3.1), $\mathbb{P}(\tau_{Bx} < \infty) \ge (M_1 Bx)^{-1}$. Fix B sufficiently large that subtracting (3.4)–(3.6) leaves a probability at least $c_*/x > 0$. On the remaining event some $\tau_x < n \le \tau_{Bx}$ has $q_n > \delta$, and $W_{n-1} \ge x/B$. Thus

$$\max_{i<n} C_i \ge \sum_{i<n} K(n-i) C_i = q_n W_{n-1} \ge \delta x/B,$$

where the first inequality uses $\sum_{i<n} K(n-i) \le 1$. Taking $x = Bt/\delta$ proves (3.3). □

Lemma 3.3. *For every $\varepsilon > 0$ there is $D_\varepsilon < \infty$ such that, for all sufficiently large t, with $L = \lceil D_\varepsilon \log t \rceil$,*

$$E_{t,L} := \{\max_{0\le n\le L} C_n \ge t\}, \qquad \mathbb{P}(E_{t,L}) \ge t^{-1-\varepsilon}. \tag{3.7}$$

Proof. Constrained concatenation gives $C_{n+m}(\omega) \ge C_n(\omega) C_m(\vartheta^n \omega)$, where $(\vartheta^n\omega)_j = \omega_{n+j}$. Choose a large fixed $v > 1$ with $c_\beta/(2v) \ge v^{-1-\varepsilon/2}$, then a finite M with $\mathbb{P}(\max_{n\le M} C_n \ge v) \ge c_\beta/(2v)$, by proposition 3.2. Repeat this success at its first hitting time, using fresh future Gaussian coordinates. The probability of k successive successes is at least $v^{-k(1+\varepsilon/2)}$, their total time is at most kM, and concatenation gives a value at least v^k. Take $k = \lceil \log_v t \rceil$. For large t the fixed extra factor is absorbed in $t^{-\varepsilon/2}$, and $kM \le D_\varepsilon \log t$ after enlarging D_ε. □

4 A positive field shift

We use a change of environment law that makes favorable stretches more likely. Sample a renewal σ with law P_τ and independent standard Gaussians (G_n), and set

$$\omega_n = G_n + \beta 1_{\{n\in\sigma\}}.$$

Denote the resulting joint law by Q. Completing the square in the Gaussian density gives, for every nonnegative measurable function f of the first N environment coordinates,

$$E_Q f(\omega) = \mathbb{E}[W_N f(\omega)], \qquad E_Q[1_{\{N\in\sigma\}} f(\omega)] = \mathbb{E}[C_N f(\omega)]. \tag{4.1}$$

These are finite-dimensional identities. For $E_{t,L}$ in lemma 3.3, stop W at the first constrained crossing of t, truncated at L. On success its stopped value is at least t. Therefore

$$p := Q(E_{t,L}) = \mathbb{E}[W_L; E_{t,L}] \geq t\mathbb{P}(E_{t,L}) \geq t^{-\varepsilon}. \tag{4.2}$$

On the original environment law, the martingale maximal inequality and $C_n \leq W_n$ give $\mathbb{P}(E_{t,L}) \leq 1/t$.

Lemma 4.1. *Let $\mathcal{A}_N$ be the event that $E_{t,L}$ occurs in some window starting at an integer in $[0, N-L]$. Then*

$$\mathbb{P}(\mathcal{A}_N) \leq (N+1)/t, \qquad \mathbb{E}[C_N; \mathcal{A}_N^c] \leq CL^{s+1}p^{-(s+2)}K(N) \quad (N \geq 2L). \tag{4.3}$$

Here C depends only on the renewal law.

Proof. The first bound is a union bound over deterministic window starts. Under Q, begin a test at the planted renewal zero, use the next L coordinates, and start the next test at $T_L = \min\{\sigma_j : \sigma_j \geq L\}$. Iterate this rule. At each start, the future renewal gaps and the Gaussian coordinates strictly after that start are fresh. The pairs consisting of the failure indicator and the next regeneration length are consequently iid. Failure may depend on that length; we retain this dependence in

$$g(n) = Q(E_{t,L}^c, T_L = n), \qquad n \geq L, \qquad \sum_n g(n) = 1 - p.$$

The test uses no Gaussian coordinate after its next regeneration, and a subsequent test excludes the coordinate at its own start. Thus even when $T_L = L$, the two tests use disjoint noise. Independence follows by conditioning on each possible renewal-gap index and physical regeneration time and applying iid independence.

The last renewal before L gives

$$g(n) \leq P_\sigma(T_L = n) = \sum_{j=0}^{L-1} u(j)K(n-j) \leq B_L K(n), \qquad B_L \leq CL^s. \tag{4.4}$$

For $n \geq 2L$ use $K(n-j) \leq 2^s K(n)$ and $\sum_{j<L} u(j) \leq L$. For $L \leq n < 2L$, the probability is at most one whereas $K(n) \geq c(2L)^{-s}$.

For any nonnegative kernel g of mass $1-p$ bounded by BK, one has

$$g^{*m}(n) \leq Bm^{s+1}(1-p)^{m-1}K(n), \qquad \sum_{m\geq 1} g^{*m}(n) \leq C_s B p^{-(s+2)}K(n). \tag{4.5}$$

Indeed one of the m summands of a composition of n is at least n/m. Its factor is at most $Bm^s K(n)$. Sum the other factors using their total mass and then sum over the m possible positions. Finally $\sum_{m\geq 1} m^{s+1}(1-p)^{m-1} \leq C_s p^{-(s+2)}$: for $p \leq 1/2$ compare with a polynomial-exponential integral, and for $p \geq 1/2$ use a convergent geometric majorant. The case $p = 1$, $g = 0$, is immediate.

On $\{N \in \sigma\} \cap \mathcal{A}_N^c$, every test starting at or before $N - L$ fails. Its next regeneration is at most N, since N itself is a renewal at least L beyond that start. The remaining distance r after the last

test therefore satisfies $0 \le r < L$. Set $h_L(r) = u(r)1_{\{0\le r<L\}}$, with g^{*0} the unit mass at zero. By regeneration and (4.1),

$$\mathbb{E}[C_N; \mathcal{A}_N^c] = Q(N \in \sigma, \mathcal{A}_N^c) \le \sum_{m\ge 0} (g^{*m} * h_L)(N).$$

For $N \ge 2L$ the $m = 0$ term vanishes. In the other terms, $N - r \ge N/2$, so $K(N - r) \le 2^s K(N)$ and $\sum_r h_L(r) \le L$. Apply (4.5) and (4.4). □

The factor $K(N)$ in (4.3) supplies the summable power needed below. It comes from retaining the planted endpoint $N \in \sigma$.

Proposition 4.2. *For some $\theta \in (4/5, 1)$,*

$$\sum_{N\ge 0} \mathbb{E}C_N^\theta < \infty. \tag{4.6}$$

Proof. Choose $\theta \in (1/s, 1)$, then $b > 1 + 1/(1-\theta)$ and $\varepsilon > 0$ so small that $\theta[s - b\varepsilon(s+2)] > 1$. Take $t = N^b$ and $L = \lceil D_\varepsilon \log t \rceil$. For all large N, $N \ge 2L$ and (4.2) gives $p \ge t^{-\varepsilon}$. Hölder's inequality on $\mathcal{A}_N$ and its complement, together with $\mathbb{E}C_N = u(N) \le 1$, yields

$$\mathbb{E}C_N^\theta \le \left(\frac{N+1}{t}\right)^{1-\theta} + \left[CL^{s+1}t^{\varepsilon(s+2)}K(N)\right]^\theta. \tag{4.7}$$

The powers of N are respectively $-(b-1)(1-\theta) < -1$ and $-\theta[s - b\varepsilon(s+2)] < -1$, with only a fixed logarithmic factor in the second term. Both series converge. The finitely many remaining moments are finite by the lognormal moment formula. □

We finish this range of beta by turning (4.6) into an actual positive change of field. Let $C_n^{(\Delta)}$ be the constrained partition at $h_a + \Delta$, and put $A_n(\Delta) = \mathbb{E}[(C_n^{(\Delta)})^\theta]$, $A_0(\Delta) = 1$.

Proposition 4.3. *Under strong disorder and $\beta > \beta_2$, there exists $\Delta > 0$ such that $F(\beta, h_a + \Delta) = 0$.*

Proof. Fix an integer $k \ge 1$. Decompose a path constrained at $N \ge k$ by its last contact $j < k$ and first contact $i \ge k$. The factors $C_j^{(\Delta)}$, $K(i-j)e^\Delta\eta_i$, and $C_{N-i}^{(\Delta)} \circ \vartheta^i$ use disjoint environment coordinates. Since $(\sum x_j)^\theta \le \sum x_j^\theta$ for nonnegative x_j,

$$\begin{aligned} A_N(\Delta) &\le \sum_{i=k}^{N} Q_i(\Delta) A_{N-i}(\Delta), \\ Q_i(\Delta) &= \mathbb{E}\eta_1^\theta\, e^{\theta\Delta} \sum_{j<k} K(i-j)^\theta A_j(\Delta). \end{aligned} \tag{4.8}$$

By proposition 4.2 and $s\theta > 1$,

$$\rho_k(0) := \sum_{i\ge k} Q_i(0) = \mathbb{E}\eta_1^\theta \sum_{j<k} A_j(0) \sum_{r\ge k-j} K(r)^\theta \longrightarrow 0.$$

For each fixed j the inner tail tends to zero and it is uniformly bounded by $\sum_r K(r)^\theta < \infty$, so dominated summation applies. Choose k with $\rho_k(0) < 1$. Since a path at j has at most j contacts, $A_j(\Delta) \le e^{\theta\Delta j} A_j(0)$, and hence

$$\rho_k(\Delta) \le e^{\theta\Delta k}\rho_k(0) < 1$$

for a sufficiently small, strictly positive Δ. Sum (4.8) only up to a finite M before rearranging:

$$\sum_{n\le M} A_n(\Delta) \le \frac{\sum_{j<k} A_j(\Delta)}{1-\rho_k(\Delta)}.$$

It follows that $\sum_n A_n(\Delta) < \infty$. The last-contact formula (2.1), with the additional reward e^{Δ} at each contact, now gives

$$\mathbb{E}Z^{\theta}_{N,h_a+\Delta} \le \sum_{j\le N} A_j(\Delta)\overline{K}(N-j)^{\theta} \le \sum_{j\ge 0} A_j(\Delta).$$

Jensen's inequality bounds $\mathbb{E}\log Z_{N,h_a+\Delta}$ uniformly from above. The no-contact paths give $\log Z_{N,h_a+\Delta} \ge \log \overline{K}(N)$, whose ratio to N tends to zero. Thus $F(\beta, h_a+\Delta) = 0$. □

5 The second-moment boundary

We must still exclude strong disorder when $0 < \beta \le \beta_2$. We use a quantitative estimate for a renewal pinned on another renewal from Alexander and Berger [1]. Its fractional-moment form is important: a statement only about a free-energy critical point would not give the infinite partition function used here.

Let σ and τ be independent renewals with gap law K. For $\lambda > 0$, define the partition constrained at the jth spine renewal by

$$V_j^{\sigma}(\lambda) = E_{\tau}\left[e^{\lambda\sum_{i=1}^{j} 1_{\{\sigma_i\in\tau\}}};\ \sigma_j \in \tau\right], \qquad V_0^{\sigma} = 1.$$

The contact at zero is excluded. Put $z = e^{\lambda} - 1$. Expansion over the selected contacts gives

$$\mathcal{V}_j^{\sigma}(\lambda) := (1-e^{-\lambda})V_j^{\sigma}(\lambda) = \sum_{m=1}^{j} z^m \sum_{0=i_0<i_1<\cdots<i_m=j} \prod_{a=1}^{m} u(\sigma_{i_a}-\sigma_{i_{a-1}}). \tag{5.1}$$

The fixed factor is necessary because the forced endpoint has weight e^{λ} in V, but weight z in the selected-contact expansion. We retain it when using the chaos estimate in [1, Section 4], whose displayed identification of the two constrained partitions omits that factor.

Tonelli's theorem gives

$$\sum_{j\ge 1} E_{\sigma}u(\sigma_j) = \sum_{n\ge 1} u(n)^2 = R.$$

Thus the reference kernel in [1, (4.2)] is $K^*(j) = R^{-1}E_{\sigma}u(\sigma_j)$, and the critical value of z is $z_0 = 1/R$, as also follows from [1, Proposition 1.3]. Write $\lambda_0 = \log(1+1/R) = \beta_2^2$.

Here is the source estimate in the notation needed for the proof. For an integer L, let $B_a = \{(a-1)L+1,\dots,aL\}$. If $I = \{1 \le b_1 < \cdots < b_{\ell} = n\}$, $b_0 = 0$, put

$$Z_I^{\sigma} = \sum_{m=1}^{nL} R^{-m} \sum_{\substack{0=i_0<i_1<\cdots<i_m=nL \\ \{\lceil i_a/L\rceil : 1\le a\le m\}=I}} \prod_{a=1}^{m} u(\sigma_{i_a}-\sigma_{i_{a-1}}). \tag{5.2}$$

This agrees with the restricted partition in [1, (4.10)]: its between-contact factor, written $K^*(b-a)w(\sigma,a,b)$ there, is exactly $R^{-1}u(\sigma_b-\sigma_a)$.

Proposition 5.1 (Alexander–Berger)**.** *Let $\zeta = 3/4$. For every $\delta > 0$ there is L_0 such that, for every integer $L \geq L_0$, every $n \geq 1$ and every such I,*

$$E_\sigma[(Z_I^\sigma)^\zeta] \leq \delta^\ell \prod_{a=1}^{\ell} (b_a - b_{a-1})^{-3/2}. \tag{5.3}$$

This is [1, (4.12)], in the explicitly cited arXiv version. Its hypotheses (1.5) hold with both slowly varying functions equal to c and both exponents equal to $1/4$. In their notation, $\alpha_* = (1 - \alpha - \widetilde{\alpha})/\widetilde{\alpha} = 2 > 1/2$ and $\zeta = (1 + \alpha_*/4)/(1 + \alpha_*/2) = 3/4$; the exponent on block gaps is $1 + \alpha_*/4 = 3/2$. The arbitrary-small-δ assertion follows there from Lemma 4.2 and (4.22), and is not restricted to a particular mass-one normalization.

Lemma 5.2. *There is $\lambda_1 > \lambda_0$ such that*

$$\sum_{j \geq 1} E_\sigma[V_j^\sigma(\lambda_1)^{3/4}] < \infty, \qquad \sum_{j \geq 1} V_j^\sigma(\lambda_1) < \infty \quad P_\sigma\text{-a.s.} \tag{5.4}$$

Proof. Choose $\delta > 0$ so the kernel $q(a) = e^\zeta \delta a^{-3/2}$ has mass $\rho < 1$. Fix an allowed L in proposition 5.1 and put $\lambda_1 = \log(1 + R^{-1} e^{1/L}) > \lambda_0$. In a term visiting ℓ blocks, at most $L\ell$ contacts are selected. Changing z_0 to $z_0 e^{1/L}$ therefore increases that term by at most e^ℓ. By the fractional-sum inequality, (5.1) and (5.3),

$$E_\sigma[V_{nL}^\sigma(\lambda_1)^\zeta] \leq (1 - e^{-\lambda_1})^{-\zeta} \sum_{\ell \geq 1} q^{*\ell}(n).$$

Summation over n bounds the total by $(1 - e^{-\lambda_1})^{-\zeta} \rho/(1 - \rho) < \infty$.

For an arbitrary spine index j, let $M = L(\lfloor j/L \rfloor + 1)$. A direct tau gap from σ_j to σ_M gives

$$V_M^\sigma(\lambda_1) \geq V_j^\sigma(\lambda_1) e^{\lambda_1} K(\sigma_M - \sigma_j).$$

The last factor uses only the next $M - j \in \{1, \dots, L\}$ spine gaps, independently of V_j. The minimum of the L positive finite expectations $E_\sigma[(e^{\lambda_1} K(\sigma_r))^\zeta]$, $1 \leq r \leq L$, is positive. Hence summability at multiples of L implies summability at every j, each multiple being used at most L times. Tonelli gives a finite sum of the ζ powers almost surely, and then a finite sum of V_j, since eventually $V_j \leq 1$ and $V_j \leq V_j^\zeta$. □

Proposition 5.3. *For every $0 < \beta \leq \beta_2$, the martingale (W_N) is uniformly integrable and $\mathbb{E} W_\infty = 1$.*

Proof. Expanding a finite product by its last selected factor and then increasing the cutoff gives

$$V_\infty^{\sigma,\mathrm{free}}(\lambda_1) := E_\tau e^{\lambda_1 \sum_{j \geq 1} 1_{\{\sigma_j \in \tau\}}} = 1 + (1 - e^{-\lambda_1}) \sum_{j \geq 1} V_j^\sigma(\lambda_1) < \infty \quad P_\sigma\text{-a.s.} \tag{5.5}$$

Indeed $\prod_{i=1}^M e^{\lambda I_i} = 1 + \sum_{j=1}^M (e^\lambda - 1) I_j \prod_{i<j} e^{\lambda I_i}$; on $I_j = 1$, the missing endpoint reward converts the coefficient to $1 - e^{-\lambda}$. Monotone convergence and lemma 5.2 apply.

Recall that under Q, $\omega = G + \beta 1_\sigma$, where G is iid standard Gaussian and independent of σ. For every $\beta^2 \leq \lambda_1$, average W_N over G as in (4.1). The Gaussian moment formula gives

$$E_G[W_N(G + \beta 1_\sigma) \mid \sigma] = E_\tau e^{\beta^2 |\tau \cap \sigma \cap [1,N]|} \leq V_\infty^{\sigma,\mathrm{free}}(\lambda_1).$$

Conditional Markov's inequality and (4.1) imply, uniformly in physical time N,

$$\mathbb{E}[W_N; W_N > A] = Q(W_N > A) \le E_\sigma \min\{1, V_\infty^{\sigma,\mathrm{free}}(\lambda_1)/A\} \longrightarrow 0 \quad (A \to \infty).$$

The last limit is dominated convergence. Thus W_N is uniformly integrable and its almost-sure limit has mean one. Since $\lambda_1 > \beta_2^2$, this covers the asserted interval, including its endpoint. □

Proof of theorem 1.1. Fix $\beta > 0$ with $\mathbb{P}(W_\infty = 0) = 1$. By proposition 5.3, $\beta > \beta_2$. Proposition 4.3 then gives a $\Delta > 0$ with $F(\beta, h_a + \Delta) = 0$. The free partition increases with h, so $h_c(\beta) \ge h_a + \Delta > h_a$. □

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OpenAI Codex was used substantially in developing the mathematical argument, checking source applications, and preparing the text. AI reader and referee-style feedback was used in revising the text. Such feedback is not independent human mathematical validation.

References

[1] K. S. Alexander and Q. Berger, *Pinning of a renewal on a quenched renewal*, arXiv:1608.03265v2, 26 April 2017. https://arxiv.org/abs/1608.03265v2.

[2] Q. Berger, *Random Interfaces and Pinning Models*, lecture notes, 4 March 2026. https://www.math.univ-paris13.fr/~quentin.berger/documents/Random-Interfaces.pdf.

[3] S. Junk and H. Lacoin, *Coincidence of critical points for directed polymers for general environments and random walks*, arXiv:2502.04113v2, 2025. https://arxiv.org/abs/2502.04113v2.

[4] H. Lacoin, *The martingale approach to disorder irrelevance for pinning models*, arXiv:1002.4753v2, 14 March 2011. https://arxiv.org/abs/1002.4753v2.

[5] F. L. Toninelli, *Disordered pinning models and copolymers: beyond annealed bounds*, Annals of Applied Probability **18** (2008), 1569–1587. https://doi.org/10.1214/07-AAP496.